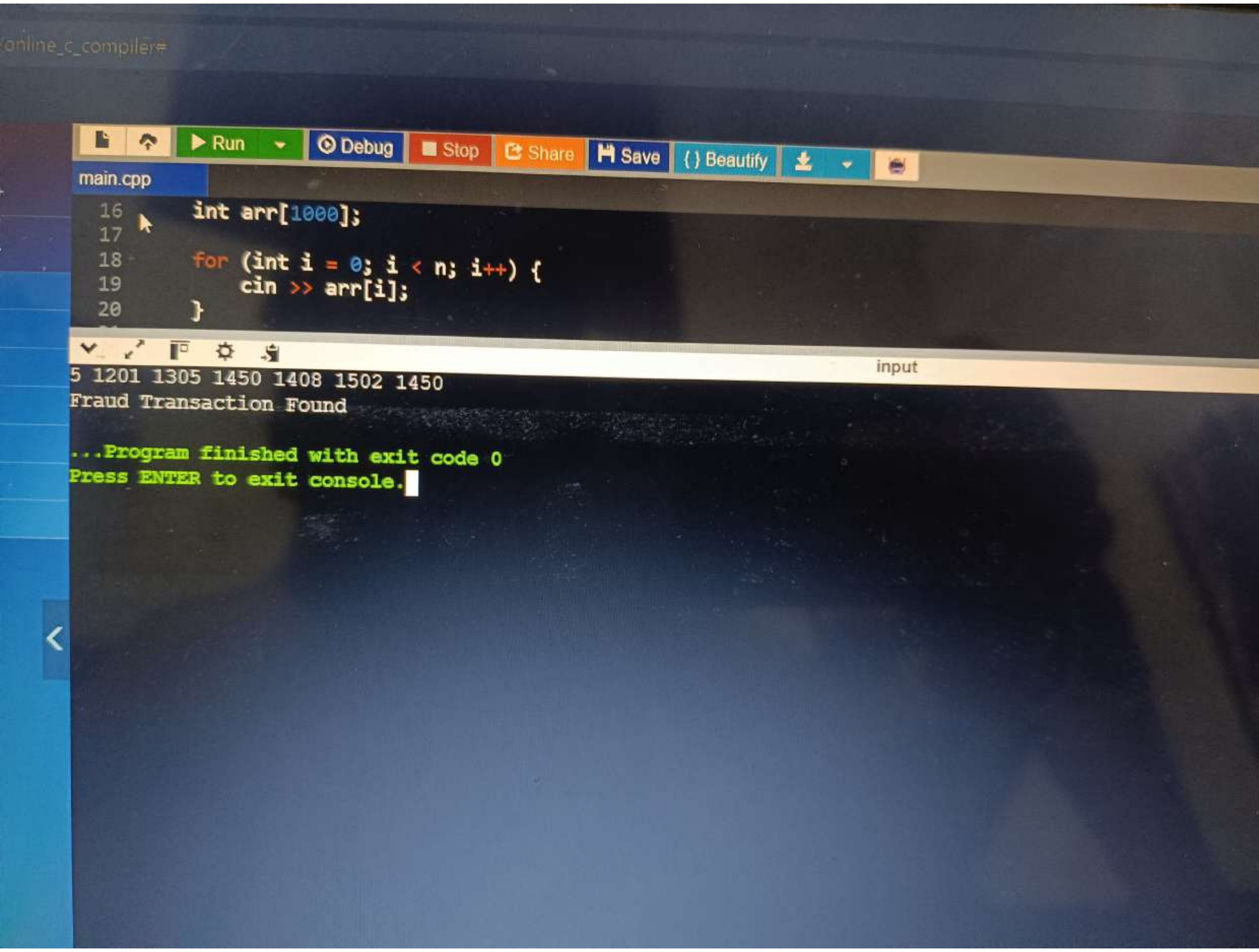
Task-2

1. Fraud Transaction Detection - Linear Search

```
#include <iostream>
using name space std;

int main() {
 int n, Target;
 cin>>n;
 int arr[n];
 for (int i=0; i<n; i++) {
   cin>> arr [i];
 }
  cin>> Target;
  bool found = false;
  for(int i=0; i<n; i++) {
   if (arr[i] = = Target) {
      found = true;
    break;
 }
}
  if (found)
   cout<<"Fraud Transaction found";
  else
   cout<<"Transaction Not found";
   return 0;
 }
```


online_c_compiler#
Run
Debug
Stop
Share
Save
{ } Beautify
main.cpp
16 int arr[1000];
17
18 for (int i = 0; i < n; i++) {
19 cin >> arr[i];
20 }
input
5 1201 1305 1450 1408 1502 1450
Fraud Transaction Found
...Program finished with exit code 0
Press ENTER to exit console.

```
#include <iostream>
using namespace std;
int main() {
  int n, ~~integer~~ target;
   cin>>n;
  int arr[n];

for (int=0; i<n; i++){
   cin>>arr[i];
}

   ~~cin>target~~
   cin>>target;
   int low = 0, high = n-1;
   bool found = false;
   while (low <= high) {
     int mid = (low+high)/2;
    if(arr[mid] == target) {
      found = True;
      break;
   } else if (arr[mid] < target) {
      low = mid+1;
    }
    else {
      high = mid-1;
    }
 }
    if (found)
       cout<<"log found";
    else
       cout<<"Log not found";
    return 0;
}
```

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main.cpp

```cpp
        int high = n - 1;
        bool found = false;

        while (low <= high) {
            int mid = (low + high) / 2;

            if (arr[mid] == target) {
                found = true;
                break;
```

input

```
7 1001 1005 1010 1015 1020 1030 1040


1015
Log Found

...Program finished with exit code 0
Press ENTER to exit console.
```

3. Sort Employee Salaries – Bubble Sort

```
#include <iostream>
using namespace std;
int main() {
  int n;
  cin>> n;
  int arr[n];
  for (int i=0; i<n; i++) {
      cin>>arr[i];
  }
   for (int i=0; i<n-1; i++) {
    for (int j=0; j<n-i-1; j++) {
      if (arr[j] > arr[j+1]) {
       int temp = arr[j];
       arr[j] = arr[j+1];
       arr[j+1] = temp;
    }
   }
}
  for (int i=0; i<n; i++) {
     cout<<arr[i]<<" ";
  }
     return 0;
  }
```

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main.cpp

```cpp
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
```

input

```
5 45000 32000 55000 28000 60000
28000 32000 45000 55000 60000

...Program finished with exit code 0
Press ENTER to exit console.
```

4. Sort website Response Time - Insertion Sort.

```
#include <iostream>
using namespace std;
int main(){
 int n;
 cin>>n;
 int arr[1000];
 for(int i=0; i<n; i++){
    ~~cin>>~~ cin >> arr[i];
 }
 for (int i=1; i<n; i++){
    int key = arr[i];
    int j=i-1;
   while (j>=0 && arr[j]>key){
      arr [j+1] = arr [j];
        j--;
   }
    arr[j+1] = key;
 };
  for (int i=0; i<n; i++){
     cout<<arr[i]<<" ";
  }
  return 0;
  }
```

c++_compiler

Run | Debug | Stop | Share | Save | { } Beautify

main.cpp

```
    cin >> n;

    int arr[1000];

    for (int i = 0; i < n; i++) {
        cin >> arr[i];
```

input

```
5 250 120 320 180 100
100 120 180 250 320

...Program finished with exit code 0
Press ENTER to exit console.
```

5. Prioritize Bug Reports - Selection Sort.

```
#include <iostream>
using namespace std;
int main(){
 int n;
 cin>>n;
 int arr[1000];
 for (int i=0; i<n; i++){
 } cin>> arr[i];
 for(int i=0; i<n-1; i++) {
   int minIndex =i;
 for (int j=i+1; j<n; j++) {
   if(arr[j] <arr[minindex]){
      minindex =j;
  }
 }
   int temp = arr[i];
   arr[i] = arr [minindex];
   arr [minIndex] = temp;
 }
 for (int i=0; i<n; i++){
   cout <<arr[i]<<" ";
 }
   return0;
}
```

Run Debug Stop Share Save { } Beautify

main.cpp

```cpp
    int arr[1000];

    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
```

input

```
5 9 2 8 1 5
1 2 5 8 9


...Program finished with exit code 0
Press ENTER to exit console.
```